

# Lab 4 Activity

## 1

Load in the `C02` data from the `datasets` package. Look at the documentation to understand the variables in this dataset. Identify a way to analyze these data using a  $t$  test (several possible choices, but `conc` probably won't be an interesting DV). Evaluate each assumption, conduct the  $t$  test, compute an effect size, and interpret the results.

## 2

Load in the `sleep` data from the `datasets` package. Look at the documentation to understand the variables in this dataset (but try not to peek at the examples section of the documentation). Identify a way to analyze these data using a  $t$  test. Evaluate each assumption, conduct the  $t$  test, compute an effect size, and interpret the results.

Hint: run the following code to transform the data into “long” format, which will make the dataset easier to work with:

```
sleep2 <- sleep %>%  
  pivot_wider(id_cols = ID,  
              names_from = group,  
              names_prefix = "group",  
              values_from = extra)
```